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$$f(x, y) = \frac{4}{\pi} \arctan(xy) \text{ in het punt } a(1, 1)$$

$$f(1, 1) = 1$$

$$\frac{\partial f}{\partial x}(x, y) = \frac{4}{\pi} \cdot \frac{1}{1 + (xy)^2} \cdot y$$

$$\frac{\partial f}{\partial x}(1, 1) = \frac{2}{\pi}$$

$$\frac{\partial f}{\partial y}(x, y) = \frac{4}{\pi} \cdot \frac{1}{1 + (xy)^2} \cdot x$$

$$\frac{\partial f}{\partial y}(1, 1) = \frac{2}{\pi}$$

$$\Rightarrow z = 1 + \frac{2}{\pi}(x - 1) + \frac{2}{\pi}(y - 1)$$

$$= \frac{2}{\pi}(x + y - 2) + 1$$

References